



中国科学技术大学

数理方程 UNIVERSITY OF SCIENCE AND TECHNOLOGY OF CHINA

第四周作业

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6. 解: 考虑定解问题:
$$\begin{cases} u_t = a^2 u_{xx} & 0 < x < 2l, t > 0 \\ u_x(0, t) = u_x(2l, t) = 0 & t > 0 \\ u(x, 0) = \varphi(x) \end{cases}$$

设 $u(x, t) = X(x)T(t)$, 代入方程有 $\frac{T'(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$

即有 $T'(t) + \lambda a^2 T(t) = 0$; $\begin{cases} X''(x) + \lambda X(x) = 0 \\ X'(0) = X'(2l) = 0 \end{cases} \Rightarrow \begin{cases} \lambda_0 = 0, X_0(x) = 1 & n=0 \\ \lambda_n = k_n^2 = (\frac{n\pi}{2l})^2, X_n(x) = \cos \frac{n\pi x}{2l} & n \geq 1 \end{cases}$

将 λ_n 代入 T 的方程可得 $T_n(t) = \begin{cases} C_0 & n=0 \\ C_n e^{-(\frac{n\pi a}{2l})^2 t} & n \geq 1 \end{cases}$

故有级数解 $u(x, t) = C_0 + \sum_{n=1}^{+\infty} C_n e^{-(\frac{n\pi a}{2l})^2 t} \cos \frac{n\pi x}{2l}$

代入 $u(x, 0) = \varphi(x) \Rightarrow C_0 + \sum_{n=1}^{+\infty} C_n \cos \frac{n\pi x}{2l} = \varphi(x)$

$\Rightarrow C_0 = \frac{\int_0^{2l} \varphi(x) dx}{\int_0^{2l} 1 dx} = \frac{1}{2l}$; $C_n = \frac{\int_0^{2l} \cos \frac{n\pi x}{2l} \varphi(x) dx}{\int_0^{2l} \cos^2 \frac{n\pi x}{2l} dx} = \frac{2}{n\pi A} \sin \frac{n\pi A}{2l} \cdot \cos \frac{n\pi}{2}$

$\Rightarrow u(x, t) = \frac{1}{2l} + \sum_{n=1}^{+\infty} \frac{2}{n\pi A} \cos \frac{n\pi}{2} \sin \frac{n\pi A}{2l} e^{-(\frac{n\pi a}{2l})^2 t} \cos \frac{n\pi x}{2l}$
或写为 $u(x, t) = \frac{1}{2l} + \sum_{k=1}^{+\infty} \frac{(-1)^k}{\pi A k} \sin \frac{k\pi A}{l} e^{-(\frac{k\pi a}{l})^2 t} \cos \frac{k\pi x}{l}$

$t \rightarrow +\infty$ 时 $u(x, t) \rightarrow \frac{1}{2l}$ ($t \rightarrow +\infty$ 时 指数项 $e^{-(\frac{n\pi a}{2l})^2 t} \rightarrow 0$)

$A \rightarrow 0$ 时 $u(x, t) \rightarrow \frac{1}{2l} + \frac{1}{l} \sum_{k=1}^{+\infty} (-1)^k e^{-(\frac{k\pi a}{l})^2 t} \cdot \cos \frac{k\pi x}{l}$ ($A \rightarrow 0$ 时, $\sin \frac{n\pi A}{2l} \sim \frac{n\pi A}{2l}$)

8. 解: 考虑定解问题:
$$\begin{cases} \Delta u = 0 \\ u(a, \theta) = T\theta(\pi - \theta) & 0 < \theta < \pi \\ u(r, 0) = u(r, \pi) = 0 & 0 < r < a \end{cases}$$

设 $u(r, \theta) = R(r)\Theta(\theta)$, 代入方程有 $r^2 \frac{R''(r)}{R(r)} + r \frac{R'(r)}{R(r)} = -\frac{\Theta''(\theta)}{\Theta(\theta)} = \lambda$

即有 $r^2 R''(r) + r R'(r) - \lambda R(r) = 0$; $\begin{cases} \Theta''(\theta) + \lambda \Theta(\theta) = 0 \\ \Theta(0) = \Theta(\pi) = 0 \end{cases} \Rightarrow \lambda_n = n^2, \Theta_n(\theta) = \sin n\theta, n \geq 1$

将 λ_n 代入 R 的方程可得 $R_n(r) = a_n r^n + b_n r^{-n}$ 由 $R(r)$ 在 $r=0$ 处有限 $\Rightarrow b_n = 0$

故有级数解: $u(r, \theta) = \sum_{n=1}^{+\infty} a_n r^n \sin n\theta$

代入 $u(a, \theta) = T\theta(\pi - \theta) \Rightarrow \sum_{n=1}^{+\infty} a_n a^n \sin n\theta = T\theta(\pi - \theta)$





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$$\Rightarrow a_n = \frac{1}{a^n} \frac{\int_0^\pi T(\theta) \sin n\theta d\theta}{\int_0^\pi \sin^2 n\theta d\theta} = \begin{cases} \frac{8T}{a^n \cdot \pi n^2} & n \text{ 为奇数} \\ 0 & n \text{ 为偶数} \end{cases}$$

$$\Rightarrow u(t, \theta) = \frac{8T}{\pi} \sum_{k=0}^{+\infty} \frac{1}{(2k+1)^3 \cdot a^{2k+1}} t^{2k+1} \cdot \sin(2k+1)\theta$$

9. 解: 设 $u(x, y) = X(x)Y(y)$ 代入方程 $u_{xx} - u_y = 0$ 得: $X''(x)Y(y) = X(x)Y'(y)$

$$\frac{X''(x)}{X(x)} = \frac{Y'(y)}{Y(y)} = -\lambda \Rightarrow \begin{cases} X''(x) + \lambda X(x) = 0 \\ Y'(y) + \lambda Y(y) = 0 \end{cases}$$

① $\lambda = 0$ 时: 只有平凡解 ② $\lambda = k^2 > 0$ 时: $Y(y) = Ae^{-ky}$, $X(x) = B \cos kx + C \sin kx$
 $\lim_{x \rightarrow +\infty} u(x, y) = 0 \Rightarrow B, C$ 均为 0. 即只有平凡解

③ $\lambda = -k^2 < 0$ 时: $Y(y) = Ae^{-ky}$ $X(x) = Be^{kx} + Ce^{-kx}$
 $\lim_{x \rightarrow +\infty} u(x, y) = 0 \Rightarrow B = 0$. $u(x, y) = AC \cdot e^{-ky - kx}$
 即解为 $u(x, y) = C_1 e^{-kx - ky}$. C_1 为任意常数, $k > 0$

10. 解: (1) $\begin{cases} u_t = a^2 u_{xx} \\ u(t, 0) = u_0, u_x(t, l) = 0 \\ u(0, x) = \varphi(x) \end{cases} \quad \text{令 } v(t, x) = u(t, x) - u_0 \quad \text{并设 } v(t, x) = T(t)X(x)$

$$\text{有 } \begin{cases} v_t = a^2 v_{xx} \\ v(t, 0) = v_x(t, l) = 0 \\ v(0, x) = \varphi(x) - u_0 \end{cases} \quad \frac{1}{a^2} \frac{T'(t)}{T(t)} = \frac{X''(x)}{X(x)} = -\lambda \Rightarrow \begin{cases} T'(t) + \lambda a^2 T(t) = 0 \\ X''(x) + \lambda X(x) = 0 \\ X(0) = 0, X'(l) = 0 \end{cases}$$

解关于 $X(x)$ 的方程有 $\lambda_n = k_n^2 = \left(\frac{2n+1}{2l}\pi\right)^2$, $X_n(x) = \sin \frac{2n+1}{2l}\pi x$ $n \geq 0$

λ_n 代入 T 的方程得到 $T_n(t) = C_n e^{-\left(\frac{2n+1}{2l}\pi a\right)^2 t}$

$$v(t, x) = \sum_{n=0}^{+\infty} C_n e^{-\left(\frac{2n+1}{2l}\pi a\right)^2 t} \sin \frac{2n+1}{2l}\pi x$$

由 $v(0, x) = \varphi(x) - u_0$ 得 $\sum_{n=0}^{+\infty} C_n \sin \frac{2n+1}{2l}\pi x = \varphi(x) - u_0$

$$\Rightarrow C_n = \frac{\int_0^l (\varphi(x) - u_0) \sin \frac{2n+1}{2l}\pi x dx}{\int_0^l \sin^2 \frac{2n+1}{2l}\pi x dx} = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{2n+1}{2l}\pi x dx - \frac{4u_0}{(2n+1)\pi}$$

$$\text{故 } u(t, x) = u_0 + \sum_{n=0}^{+\infty} \left[\frac{2}{l} \int_0^l \varphi(x) \sin \frac{2n+1}{2l}\pi x dx - \frac{4u_0}{(2n+1)\pi} \right] e^{-\left(\frac{2n+1}{2l}\pi a\right)^2 t} \sin \frac{2n+1}{2l}\pi x$$





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$$(4) \begin{cases} u_{tt} = a^2 u_{xx} + b \sinh x \\ u(t, 0) = u(t, l) = 0 \\ u(0, x) = u_t(0, x) = 0 \end{cases} \quad \text{设 } u(t, x) = v(t, x) + w(x)$$

$$\text{其中 } \begin{cases} a^2 w''(x) + b \sinh x = 0 \\ w(0) = w(l) = 0 \end{cases} \quad \begin{cases} v_{tt} = a^2 v_{xx} \\ v(t, 0) = v(t, l) = 0 \\ v_t(0, x) = 0 \\ v(0, x) = -w(x) \end{cases}$$

求解关于 $w(x)$ 的方程得到 $w(x) = -\frac{b}{a^2} \sinh x + \frac{b}{a^2 l} \sinh l \cdot x$

设 $v(t, x) = T(t)X(x)$, 有 $\frac{T''(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$, $X(0) = X(l) = 0$

$$\Rightarrow \lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin \frac{n\pi x}{l}, \quad n \geq 1$$

$$\text{代入 } T \text{ 方程得 } T_n(t) = a_n \cos\left(\frac{n\pi a t}{l}\right) + b_n \sin\left(\frac{n\pi a t}{l}\right)$$

$$v_t(0, x) = 0 \Rightarrow b_n = 0 \quad \text{即有 } T_n(t) = a_n \cos\left(\frac{n\pi a t}{l}\right)$$

$$v(0, x) = \sum_{n=1}^{+\infty} a_n \sin \frac{n\pi x}{l} = -w(x) \Rightarrow a_n = \frac{1}{\int_0^l \sin^2 \frac{n\pi x}{l} dx} \int_0^l (-w(x)) \sin \frac{n\pi x}{l} dx$$

$$\text{求得 } a_n = \frac{2bl^2(-1)^n \sinh l}{a^2 n\pi (n^2\pi^2 + l^2)}$$

$$\text{故 } u(t, x) = w(x) + \sum_{n=1}^{+\infty} a_n \cos \frac{n\pi a t}{l} \cdot \sin \frac{n\pi x}{l}$$

$$\text{其中 } w(x) = -\frac{b}{a^2} \sinh x + \frac{b}{a^2 l} \sinh l \cdot x, \quad a_n = \frac{2bl^2(-1)^n \sinh l}{a^2 n\pi (n^2\pi^2 + l^2)}$$

$$5. (3) \begin{cases} u_{tt} = a^2 u_{xx} - zh u_t \\ u(t, 0) = u(t, l) = 0 \\ u(0, x) = \varphi(x), \quad u_t(0, x) = \psi(x) \end{cases}$$

$$\text{令 } u(t, x) = T(t)X(x), \text{ 有 } \frac{T''(t)}{a^2 T(t)} + \frac{zhT'(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

$$X(0) = X(l) = 0$$

$$\Rightarrow \lambda_n = \left(\frac{n\pi}{l}\right)^2, \quad X_n(x) = \sin \frac{n\pi x}{l}, \quad n \geq 1$$

$$\text{代入 } T \text{ 的方程: } T''(t) + zhT'(t) + \lambda_n a^2 T(t) = 0$$

$$\text{解得 } T_n(t) = e^{-ht} (a_n \cos \omega_n t + b_n \sin \omega_n t), \quad \text{其中 } \omega_n = \sqrt{\frac{n^2\pi^2 a^2}{l^2} - h^2}$$

$$u(0, x) = \sum_{n=1}^{+\infty} a_n \sin \frac{n\pi x}{l} = \varphi(x) \Rightarrow a_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} dx$$

$$u_t(0, x) = \sum_{n=1}^{+\infty} (\omega_n b_n - h a_n) \sin \frac{n\pi x}{l} = \psi(x) \Rightarrow \omega_n b_n - h a_n = \frac{2}{l} \int_0^l \psi(x) \sin \frac{n\pi x}{l} dx$$

$$\text{代入得 } u(t, x) = \sum_{n=1}^{+\infty} e^{-ht} (a_n \cos \omega_n t + b_n \sin \omega_n t) \sin \frac{n\pi x}{l}$$

$$\text{其中 } \omega_n = \sqrt{\frac{n^2\pi^2 a^2}{l^2} - h^2}, \quad a_n = \frac{2}{l} \int_0^l \varphi(x) \sin \frac{n\pi x}{l} dx, \quad b_n = \frac{a_n}{\omega_n} h + \frac{2}{\omega_n l} \int_0^l \psi(x) \sin \frac{n\pi x}{l} dx$$





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$$(4) \begin{cases} u_{tt} = a^2 u_{xx} \\ u_x(t, 0) = 0, u_x(t, l) + h u(t, l) = 0 \\ u(0, x) = \varphi(x), u_t(0, x) = \psi(x) \end{cases} \quad \text{设 } u(t, x) = T(t)X(x), \quad \frac{T''(t)}{a^2 T(t)} = \frac{X''(x)}{X(x)} = -\lambda$$

$$X'(0) = 0, X'(l) + hX(l) = 0$$

得 $\lambda_n = k_n^2$, 其中 $\tan k_n l = \frac{h}{k_n}$ $X_n = \cos k_n x, n \geq 1$

$$T''(t) + \lambda_n a^2 T(t) = 0 \Rightarrow T_n(t) = a_n \cos k_n a t + b_n \sin k_n a t$$

$$u(0, x) = \varphi(x) \Rightarrow a_n = \frac{1}{\int_0^l \cos^2 k_n x dx} \int_0^l \varphi(x) \cos k_n x dx$$

其中 $\int_0^l \cos^2 k_n x dx = \frac{l}{2} [1 + \frac{h}{l(h^2 + k_n^2)}]$

$$u_t(0, x) = \psi(x) \Rightarrow b_n = \frac{1}{k_n \int_0^l \cos^2 k_n x dx} \int_0^l \psi(x) \cos k_n x dx$$

$$u(t, x) = \sum_{n=1}^{+\infty} (a_n \cos k_n a t + b_n \sin k_n a t) \cos k_n x$$

$$(6) \begin{cases} \Delta u = 0 & a < r < b \\ u(a, \theta) = 1, u(b, \theta) = 0 \end{cases} \quad \text{设 } u(r, \theta) = A_0 + B_0 \ln r + \sum_{k=1}^{\infty} (A_k r^k + B_k r^{-k}) (C_k \cos k\theta + D_k \sin k\theta)$$

$$\text{由 } u(a, \theta) = 1, u(b, \theta) = 0 \Rightarrow A_k, B_k = 0 \quad ; \quad \begin{cases} A_0 + B_0 \ln a = 1 \\ A_0 + B_0 \ln b = 0 \end{cases} \Rightarrow \begin{cases} A_0 = \frac{\ln b}{\ln b - \ln a} \\ B_0 = \frac{1}{\ln a - \ln b} \end{cases}$$

$$\text{故 } u(r, \theta) = \frac{\ln b - \ln r}{\ln b - \ln a} \quad a < r < b$$



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